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MILITARY SPECIFICATION

CARRIERS, TRACKED; 6 TON: M548, M548A1 AND M730A1 (OVERHAUL)

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the overhaul of the unarmored, full-tracked vehicles designed as cargo, ammunition support or guided missile equipment carriers, for cross-country and amphibious operations (see 6.1). The word "vehicles" in this specification shall imply overhauled vehicles.

1.2 Classification. Vehicles shall be classified as follows:

Type I	- Cargo and ammunition carrier, M548
Type II	- Cargo and ammunition carrier, M548A1
Type III	- Guided missile equipment carrier, M730A1

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-S-12158	- Sealing Compound, Non-curing, Polybutene.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-C-62015	- Carrier, Cargo, Full-tracked, M548, Processing for Storage and Shipment of.
MIL-C-62053	- Carrier, Guided Missile Equipment, M730 and M730A1; Processing for Storage and Shipment of.
MIL-T-62099	- Transmission, Hydraulic: Modified Automatic TX-100-1.
MIL-E-62140	- Engine, Diesel: Liquid Cooled, V-type, 6 Cylinder, 210 Hp.

STANDARDS
FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.

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MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-1275	- Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.
MIL-STD-1474	- Noise Limits for Army Materiel.
MIL-STD-45662	- Calibration Systems Requirements.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise speciifed, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

8736607	- Carrier, Cargo, Tracked; 6 Ton, M548.
8750029	- Carrier, Cargo, Tracked; 6 Ton, M548A1.
8750067	- Carrier, Guided Missile Equipment, M730A1.
10942803	- Lubrication Chart, Production, Fuel and Coolant.

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish vehicles which shall be subjected to first article inspection (see 4.3). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent vehicles delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Material used in the overhaul of the vehicle shall be as specified herein and in referenced drawings, specifications and standards. Material shall be free of defects and imperfections that might affect the serviceability and reliability of the finished product (see 4.8.1 and 6.9).

3.3 Design and construction. Completed vehicles shall conform to this specification and the Army technical data packages as follows (see 4.8.1):

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Type I	- 8736607
Type II	- 8750029
Type III	- 8750067

3.3.1 Parts and components. Components, assemblies and subassemblies requiring overhaul shall be overhauled in accordance with the contract and this specification. When new parts are required they shall conform to specifications, standards and drawings listed in the respective vehicle technical data package (see 4.8.1).

3.3.2 Qualified products list (QPL). The contractor shall be responsible for using parts and assemblies necessary in vehicle overhaul that are listed as qualified products on the QPL and those approved for inclusion on the QPL. The contractor's manufacturing inspection records shall specifically list all QPL items by part or drawing number(s), name of supplier(s), and number and date of the QPL. When assemblies and parts are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document with the name of supplier(s) (see 4.1.4 and 4.8.1).

3.3.3 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications or drawings (see 4.8.1).

3.3.4 Break-in run. Prior to being subjected to testing, each vehicle shall be given a break-in run (see 4.8.3). The break-in run is not intended to be a part of any test but is carried out by the contractor to insure that the vehicle is in a safe, operable condition prior to testing. A rated payload condition is not required. When specified (see 6.2), the break-in run shall be simulated (see 3.3.4.3.2 and 4.8.3.1).

3.3.4.1 Preparation. Prior to the break-in run, except for equipment and component installation and servicing performed by the contractor as a final processing of vehicles for acceptance, the vehicle shall be completely assembled and all adjustments completed. The vehicle, including all components requiring lubrication, shall be lubricated and serviced with fuel conforming to VV-F-800. Lubricants and fluids shall be as specified in the production lubrication drawing 10942803. Specific application by grades or types for various expected seasonal temperature ranges shall be as specified (see 4.8.2).

3.3.4.2 Oil pressure and temperature. Prior to the start of the break-in run, the engine shall be operated at 500 to 600 revolutions per minute (rpm), until engine temperature and oil pressure stabilize. If, at anytime during the break-in run, the engine has been stopped for more than 15 minutes, the engine shall again be operated, as above, before continuing the break-in run. Proper oil pressure and temperature shall be maintained during operation on level ground, 60 percent (%) grades and 30% side slopes.

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3.3.4.3 Operation and distance. Prior to tests, each overhauled vehicle shall be given a break-in run for the distances specified in table I on smooth, level, hard-surfaced roads, or test track. The vehicle shall be operated over the undulating section of road (see figure 1) during a. and b. divisions of the break-in run, twice in each direction.

TABLE I. Speeds and distances for break-in run.

Division of run	Speed Miles per hour (mph)	Distance Miles	Test conditions
a.	0 to 10	5	Track (road)
b.	11 to 20	5	Track (road)
c.	21 to 30	10	Track (road)
d.	31 to maximum	10	Track (road)

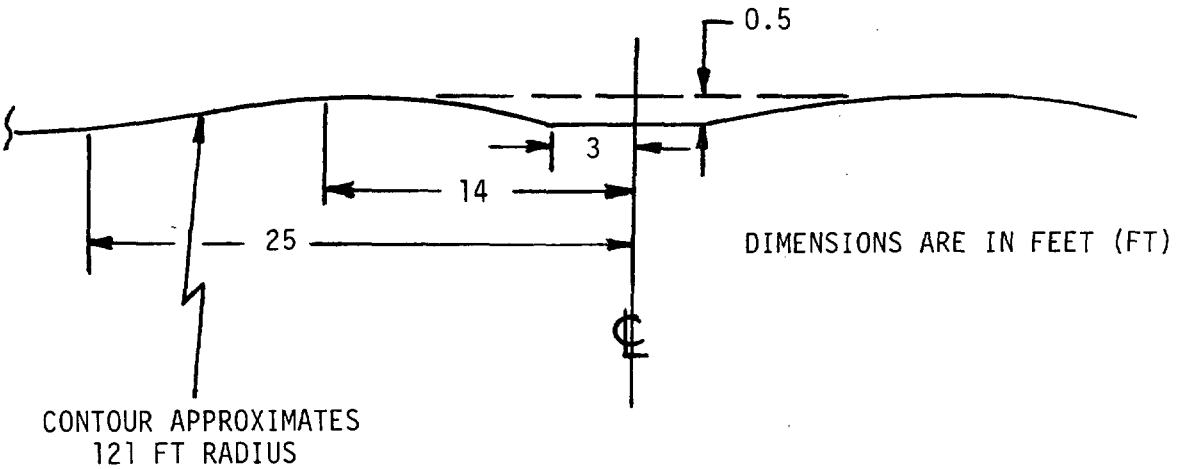


FIGURE 1. Profile of test track (artificial bump).

3.3.4.3.1 Reverse operation. After each division of the break-in run, the vehicle shall be stopped, the engine allowed to idle for not less than 2 minutes, and the vehicle driven in reverse for a distance of not less than 50 feet (ft).

3.3.4.3.2 Break-in run (simulated). When specified (see 6.2), prior to test, each vehicle shall be given a break-in run on the Army skid-type dynamometer. The break-in run shall be in accordance with table II and shall meet the requirements specified herein (see 4.8.3.1).

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TABLE II. Simulated break-in run.

Division of run	Speed Miles per hour (mph)	Distance Miles	Test conditions
a.	0 to 10	5	Dynamometer
b.	11 to 20	5	Dynamometer
c.	21 to 30	5	Track (road)
d.	31 to maximum	5	Track (road)

3.3.4.3.3 Condition after run. After completion of the break-in run, the engine shall run smoothly. Prior to vehicle being submitted to the government for acceptance, all suspension wheel bearings shall be examined to determine if adjustments are correct, and readjusted if required. The differential oil filter shall be removed, cleaned and reinstalled following completion of the break-in run and prior to shipment (see 4.8.3.2).

3.3.5 Reliability. Reliability shall be not less than 675 Mean Miles between failure (MMBF) during the first 2000 miles when operated in accordance with the operational profile (see 3.3.7.1). For calculating MMBF, the definition in the M113A2 family 50 mile mission failure scoring criteria shall be used (see 4.8.4 and 6.9).

3.3.6 Maintainability. Total maintenance, excluding driver/crew checks and services, shall not exceed 70 manhours during 2000 miles of operation in accordance with the operational profile (see 3.3.7.1). This equates to a Maintenance Ratio (MR) of 0.35 at 10 operational miles, equivalent to 1 operational hour. The scheduled maintenance interval shall be 3 months or 750 miles, whichever comes first (see 4.8.5).

3.3.7 Durability. The vehicle, combat loaded, shall have a 50% probability of completing the first 2000 test miles without replacement, rebuild or major overhaul of the engine, transmission, transfer gear unit, differential steer unit or final drives, and without repair or rebuild of the hull, when operated in accordance with the operational profile specified in 3.3.7.1 (see 4.8.6).

3.3.7.1 Operational profile. Mileage shall be apportioned as follows:

a. 20% hard-surfaced roads.
b. 40% secondary roads.
c. 40% cross country.

Note: When vehicle is to be operated as a prime mover (equipped with air brake kit), mileage during operation on hard-surfaced roads and cross country shall include 200 miles in each mode towing a 14 000 pound (lb) gross vehicle weight (gvw) trailer for type II only.

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3.3.8 Sealing and seals.

3.3.8.1 Sealer. The sealer conforming to MIL-S-12158 shall restrict entrance of water in excess of 3 quarts per minute into the hull when the vehicle is fording streams or rivers and when operating in deep water (see 4.8.7 and 4.8.26.10).

3.3.8.2 Bearing seals. When fording, operating amphibiously (type III only), or in mud, sand or snow, bearings seals shall prevent the entrance of foreign matter into the bearings, which are exposed to contamination during these operations. Bearing seals shall restrict the leaking of lubricants from bearings (see 4.8.7).

3.3.9 Cab and tailgate seals. Cab and tailgate seals shall restrict entrance of water into the vehicle in excess of 3 quarts per minute (see 4.8.2, 4.8.7, and 4.8.26.10).

3.3.10 Special kits. When specified (see 6.2), special kits received with the vehicle shall be overhauled (see 4.8.8).

3.3.10.1 Material handling kit (types I and II). When installed, the material handling kit shall be capable of loading material weighing up to 1500 lb into the cargo area. This kit shall be mountable either left of center, right of center, or in a central location of the cargo area, without interference (see 4.8.8).

3.3.10.2 Machine gun kit (types I and II) . When mounted on the vehicle, the machine gun kit shall support the machine gun when firing, and allow the gun to rotate 360 degrees (°) to the right and left, without interference or binding (see 4.8.8).

3.3.10.3 Winterization equipment (types I, II and III). Provisions shall be made for installation of winterization equipment, including personnel heaters. When installed this winterization equipment shall assure operation under arctic conditions in ambient temperatures of minus 25 degrees Fahrenheit (°F) to minus 65°F (see 4.8.8).

3.3.10.4 Winterization kit (type III). When installed, the winterization kit shall supplement the personnel heater by providing insulation for the vehicle cab, frost shields for the cab door windows, and an insulated cab cover assembly to permit vehicle operation in ambient temperatures from minus 25°F to minus 65°F. The insulated soft cab cover shall resist the entrance of water, dust and snow. When installed, the engine coolant heater system shall assure engine starting and operation in ambient air temperature from minus 25°F to minus 65°F. The engine coolant heater shall be capable of operating for not less than 12 hours on battery power alone without discharging the battery to a level that would preclude satisfactory starting of the engine (see 4.8.8).

3.3.10.5 Driver's cab cover kit (types I, II and III). When the driver's cab cover kit is installed, it shall provide added protection for the crew during arctic operations in ambient air temperature down to minus 65°F. When installed there shall be no interference or misalignment (see 4.8.8).

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3.3.10.6 Personnel heater kit (primary) (types I, II and III). Provisions shall be made for subsequent installation of the personnel heater kit without interference with other components. When installed, this kit shall provide heat for defrosting of the windshield, and added heat in the cab for vehicle personnel spaces in ambient temperatures down to minus 25°F (see 4.8.8).

3.3.10.7 Personnel heater kit (secondary) (types I, II and III). Provisions shall be made for installation of the personnel heater and defroster kit (secondary), to be used in conjunction with the (primary) cab personnel heater and defroster kit. The kit shall provide personnel heat and windshield defrosting in ambient air temperatures down to minus 65°F. Installation of the insulated hard top cab shall be without interference or misalignment and be water, dust and snow proof (see 4.8.8).

3.3.10.8 Insulated cargo cover kit (types I and II). When installed, the insulated cargo cover kit shall provide adequate protection for personnel and material in the cargo area, during arctic operations in ambient air temperature down to minus 65°F. As installed, the kit shall be water, dust, and snow proof (see 4.8.8).

3.3.10.9 Cargo heater kit (primary)(types I and II). Provisions shall be made for the installation of the cargo heater kit (primary) without interference with other components. When installed, this kit shall provide heat for personnel seated in the rear cargo area in ambient air temperatures down to minus 25°F (see 4.8.8).

3.3.10.10 Cargo heater kit (secondary) (types I and II). Provisions shall be made for the installation of the cargo heater kit (secondary) to be used in conjunction with the cargo heater kit (primary). When installed, the insulated cargo cover shall provide protection for personnel and material in the cargo area during arctic operations in ambient air temperatures down to minus 65°F. When installed, the insulated cargo cover shall be water, dust and snow proof (see 4.8.8).

3.3.10.11 Engine stand-by coolant heater kit (types I, II and III). Provisions shall be made for installation of the engine stand-by coolant heater kit, without interference or misalignment. When installed, this kit shall, by circulating a sufficient quantity of heat through the battery heating pad and engine cooling system, assure engine starting and operation in ambient air temperatures from minus 25°F to minus 65°F. The heater shall operate for not less than 12 hours without discharging the battery sufficiently to preclude satisfactory engine starting (see 4.8.8).

3.3.10.12 Air brake kit (types I and II). Provisions shall be made for installation of the air brake kit, for use when the vehicle is required to tow other vehicles having air brakes (see 4.8.8).

3.3.11 Doors, covers and windows. All doors, covers and windows furnished shall conform to applicable drawings. When doors and windows are installed in accordance with the drawings, they shall fit and be water, dust and snow proof, and function without interference or misalignment (see 4.8.2 and 4.8.9).

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3.3.12 Engine. The engine shall conform to MIL-E-62140. The engine in the vehicle shall meet all specified vehicle tests without failure (see 4.8.2 and 4.8.10).

3.3.12.1 Engine air induction system. The air induction system on the vehicle shall be complete in accordance with applicable drawings and shall be properly installed (see 4.8.11).

3.3.12.2 Exhaust system. The exhaust system shall conform to applicable drawings and be properly installed. The concentration of carbon monoxide (CO) in occupied portions of the vehicle shall not exceed 50 parts per million (ppm), and concentrations of oxides of nitrogen (NO) shall not be more than 5 ppm. The toxic fume limitation considers operation of heaters, when installed (see 4.8.12).

3.3.13 Fuel system. During vehicle operation, the fuel system shall maintain a continuous supply of fuel to the engine, when ascending and descending a 60% grade in forward and reverse gear, and when vehicle is being operated on 30% side slopes, with left, and then right side of vehicle up slope (see 4.8.2 and 4.8.13).

3.3.13.1 Fuel system hardware. Provisions shall be made to assure the internal cleanliness of the fuel, tanks, hydraulic lines and connections prior to initial servicing of the vehicle. The fuel tanks, hydraulic reservoir, lines and connections shall be free of leakage. The fuel tank shall be capable of receiving 50 gallons per minute (gpm) (see 4.8.2 and 4.8.14).

3.3.13.2 Fuel shut-off valve. When the manual fuel shut-off handle is placed in the "off" position, the engine shall stop within 5 minutes (see 4.8.15).

3.3.13.3 Rack linkage. With rack linkage installed in accordance with applicable drawings, and with the accelerator pushed against the stop, the rack shall be at full open position (see 4.8.16).

3.3.14 Transmission, transfer gearcase, and controlled differential. The transmission shall conform to MIL-T-62099. When installed in the vehicle, the transmission and controlled differential shall operate throughout all gear ranges without evidence of leakage (see 4.8.2 and 4.8.17).

3.3.15 Electrical system.

3.3.15.1 Generating system voltage (alternator). After no less than 20 minutes of operation with headlights on, regulator control shall be observed. The alternator shall be operated at 4000 revolutions per minute (rpm) $\pm$ 100 rpm with an output of 50 amperes (A) $\pm$ 5 A, and the voltage regulator rheostat adjusted to provide 28 volts (V) direct current (dc) at the vehicle batteries. Voltage at the vehicle batteries shall not increase more than 0.5 V dc when the load is removed (see 4.8.18.1).

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3.3.15.2 Lights. Internal and external lights specified on applicable drawings shall operate, and all electrical contacts and connections shall maintain positive contact, under all vehicle operating conditions (see 4.8.18.2).

3.3.15.3 Communication and electrical circuits. Communication and electrical circuits shall maintain continuity from end to end, without shorts from pin to pin or from pin to connector shell (see 4.8.2 and 4.8.18.3).

3.3.15.4 System voltage characteristics. Electrical system voltage characteristics shall fall within the limits for transient and steady state voltage requirements specified in MIL-STD-1275 (see 4.8.18.4).

3.3.15.5 Electromagnetic interference suppression. The vehicle, when operating, shall meet the requirements for electromagnetic interference/ electromagnetic characteristics (EMI/EMC) suppression in accordance with parts 4 and 8 of MIL-STD-461 (see 4.8.18.5 and 6.7).

3.3.15.6 Heaters. The personnel and cargo area heaters shall start and operate satisfactorily on 20 V dc, using diesel fuel conforming to VV-F-800 (see 4.8.18.6).

3.3.16 Controls. All electrical, mechanical and hydraulic controls shall operate without malfunction throughout all ranges of operation under all vehicle operating conditions (see 4.8.2 and 4.8.19).

3.3.17 Adjustment mechanisms. All adjustment mechanisms shall function properly and maintain adjustment within required settings during all vehicle operating conditions (see 4.8.2 and 4.8.20).

3.3.18 Stowed material. All interior and exterior basic issue items (BII) when stowed on the vehicle in the spaces provided shall fit properly and not interfere with the operation of the vehicle and components in any manner (see 4.8.21).

3.3.19 Welding repairs. Welding repairs of any kind shall be made only when, and to the extent, specifically authorized by the procuring activity (see 4.8.22 and 6.2).

3.3.20 Folding blast covers (Type III). The folding blast covers shall provide protection for the crew compartment when launching a missile. The blast covers shall fold open and remain open to permit vehicle operation without the covers closing on the operator (see 4.8.23).

3.3.21 Tailgate (Type III). The tailgate shall hinge downward and support the weight of two men during manual missile loading of the launcher. When closed the tailgate shall be sealed to restrict the entrance of water into the vehicle during fording and flotation operations (see 4.8.24).

3.3.22 Fire extinguisher. The fixed fire extinguisher system, when installed in accordance with applicable drawings, shall be readily accessible for servicing and operation internally and externally. Maximum effort for externally discharging the system shall not exceed the value specified on the applicable drawing (see 4.8.2 and 4.8.25).

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3.4 Performance. A complete overhauled vehicle, with rated payload, and completely serviced, shall perform as specified herein. Ammunition may be simulated in size and weight. The vehicle, serviced (see 3.3.4.1) and equipped for existing climatic conditions, shall operate as specified with, or without, special equipment. The vehicle shall operate on 60% longitudinal grades, headed up and headed down grade, and on 30% side slopes, with alternate sides of vehicle up-slope. Performance shall be demonstrated on smooth, level, hard-surfaced roads free of loose material, except a section of road shall be as shown in figure 1 (see 4.8.26).

3.4.1 Environmental conditions. The complete vehicle with rated payload shall operate in ambient air temperature of plus 125°F to minus 25°F and, with special equipment installed, to minus 65°F. The complete vehicle shall withstand storage at minus 80°F without deterioration that may cause failure of any component part of the vehicle (see 4.8.26.1).

3.4.2 Power package and power train. The power package and power train, installed in accordance with overhaul technical data package, shall function smoothly throughout all gear and speed ranges without loss of lubricants. There shall be no grabbing and chattering or slippage when controls are applied to start, stop and hold the vehicle (see 4.8.26.2).

3.4.3 Cooling systems.

3.4.3.1 Engine cooling. With the vehicle operating in an ambient temperature of 125°F, the engine cooling system shall maintain engine coolant at not more than 230°F, measured at a point in the engine outlet to radiator. The radiator pressure cap shall relieve between 13 and 18 pounds per square inch gage (psig) (see 4.8.2 and 4.8.26.3).

3.4.3.2 Transmission lubricant cooling. With the vehicle operating in an ambient temperature of 125°F, the transmission lubricant cooling system shall maintain lubricant temperature at not more than 325°F, measured out of transmission into cooler, except at torque converter speed ratios less than four to ten (see 4.8.26.3).

3.4.3.3 Differential oil cooler. With the vehicle operating in an ambient temperature of 125°F, the differential lubricant cooler shall maintain lubricant temperature below 300°F measured out of differential into cooler (see 4.8.26.3).

3.4.4 Bilge pumps. The bilge pump(s), installed in accordance with applicable drawings, shall remove water from the vehicle at a rate not less than 35 gpm each with a discharge head of 4.5 ft, and with pump motor operating on 27.5 V dc impressed at the motor (see 4.8.2 and 4.8.26.4).

3.4.5 Speeds and vehicle control. The vehicle shall operate at sustained speeds up to 32 mph, and a maximum speed of 38 mph, without damage to power plant or suspension system. Performance shall be demonstrated on smooth, level, hard-surfaced terrain. With vehicle traveling between 15 mph and 20 mph at a constant speed, drift shall not exceed 4 ft in 100 ft (see 4.8.26.5).

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3.4.6 Acceleration. With the vehicle at a standstill, transmission in the 1-3 range and engine at idle speed, the vehicle shall accelerate from 0 to 20 mph in not more than 18 seconds with rated payload or 11 seconds without load on a smooth, level, hard-surfaced road (4.8.26.6).

3.4.7 Stopping. The vehicle shall stop within a distance of 40 ft from the point of service brake application when traveling at a speed of 20 mph on smooth, level, hard-surfaced roads free of loose material. Drift shall not exceed 3 ft in the stopping distance. During stopping, under the driver's control, steering action is permitted (see 4.8.26.7).

3.4.8 Turning. The vehicle, on a smooth, level, hard-surfaced road or track, with differential steering, shall make continuous, full 360° turns to the right and left, with the rear point of the outer edge of the vehicle at all times in contact with or within a 52 ft diameter circle. When in pivot steer, the vehicle shall pivot within a 30 ft diameter circle. The applied effort to the steering lever shall not exceed 100 lb to pivot the vehicle (see 4.8.26.8).

3.4.9 Water operation.

3.4.9.1 Shallow water fording. The vehicle shall ford shallow, hard-bottom salt or fresh water channels up to a depth of 40 inches. Leakage of water into the hull shall not exceed 3 quarts per minute (see 4.8.26.9).

3.4.9.2 Amphibious operation (Type III, only). With flotation kit installed, the Type III vehicle shall negotiate water of unlimited depth with the height of waves not exceeding 1 ft. When afloat in calm water not less than 15 ft deep, a maximum speed of not less than 3 mph shall be attained. The vehicle shall be capable of operating within a 30 ft corridor when crossing calm, open water. Leakage of water into the hull shall not exceed 3 quarts per minute (see 4.8.26.10).

NOTE: The types I and II carriers do not float. These vehicles must not enter water deeper than 40 inches.

3.4.10 Trench crossing. The vehicle shall cross trenches up to 36 inches in depth, and up to 66 inches in width, without damage to the vehicle (see 4.8.26.11).

3.4.11 Vertical obstacles. The vehicle, operating at low speeds in forward gear, shall cross vertical obstacles (wall) up to 24 inches in height and, in reverse gear, obstacles up to 14 inches in height without damage to the vehicle (see 4.8.26.12).

3.4.12 Grade and slope operation. During all grade and slope operation proper oil pressure and temperture shall be maintained in the engine, transmission, transfer gear unit and controlled differential. Oil pressure observation shall be made with engine operating at 600 to 800 rpm. For these operations the vehicle shall be loaded with rated payload (see 4.8.26.13 and 4.8.26.14).

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3.4.12.1 Grade operation. The vehicle shall ascend 60% grades in forward and reverse gear from a standstill position on the grade without stalling or damage to the power plant or power train. With the vehicle parked in the following positions, the engine shall be capable of restarting within 1 minute after being stopped for not less than 1 minute and after being operated for not less than 2 minutes under no load between 750 and 1000 rpm (see 4.8.26.13):

a. Headed upgrade on a 60% grade.
b. Headed downgrade on a 60% grade.

3.4.12.2 Side slope operation. The vehicle shall operate without malfunction or damage on 30% side slopes. With the vehicle parked in the following positions, the engine shall be capable of restarting within 1 minute after being stopped for not less than 1 minute and after being operated for not less than 2 minutes under no load between 750 and 1000 rpm (see 4.8.26.14):

a. Left side up on a 30% slope.
b. Right side up on a 30% slope.

3.4.13 Suspension. The vehicle suspension shall operate over a minimum of three 6 by 6 inch obstacles, spaced 30 ft apart, at a speed of 18 to 22 mph without bottoming out of the track drive sprockets, raodarm bump stops, or idler wheels, with either side or both tracks engaging the obstacles (see 4.8.2 and 4.8.26.15).

3.4.14 Winch system. The winch system shall withstand an increasing load with an 11 to 12 ft cable (total length), including three to four wraps of cable on the bare drum. The load shall be gradually increased until the shear pin fails or the load reaches 24 000 lb. The load must exceed 14 000 lb. Failure load shall be recorded (see 4.8.26.16).

3.4.14.1 Winch operation. Starting with the fifth to tenth wrap on the bare drum, tension shall be applied to provide even distribution of the cable until the cable has been completely reeled in (see 4.8.26.16).

3.4.14.2 Winch structural integrity. The winch system structural integrity shall be verified by pulling the winch cable with a 20 000 lb load. During the test, the engine power takeoff shall be disengaged. Three to six wraps of cable shall be wrapped on the bare drum. The rope length is optional (see 4.8.26.16).

3.4.14.3 Winch controls. The cab-mounted controls shall engage the power takeoff drive for operation of the winch. The manually operated device furnished shall permit free spooling of the winch drum (see 4.8.26.16).

3.4.15 Noise level. A sound level meter shall be located 4 inches forward of the engine compartment rear bulkhead, 4 inches to the left of the engine compartment left bulkhead, and 12 inches below the inside surface of the top plate. The vehicle shall travel at 30 mph in the 1-3 gear range on a hard-surfaced straight road with all doors, hatches and windows closed. With these conditions the sound level shall not exceed 119 decibels ("A" weighted system response) [dB(A)] (see 4.8.26.17).

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3.4.15.1 Noise hazard caution sign. The vehicle driver and passenger compartments shall be posted as noise hazardous areas as specified in MIL-STD-1474 (see 4.8.26.17).

3.5 Painting. Exterior and interior of vehicle and component assemblies and parts which require painting shall be prepared and painted in accordance with MIL-STD-193, MIL-E-52798 or MIL-E-52835. Exterior color shall be forest green (see 4.8.2).

3.6 Identification marking. Vehicle marking shall be in accordance with MIL-STD-642. Parts and components shall be marked and identified in accordance with MIL-STD-130. A 3-inch star shall be centered on the vehicle front and rear. Registration numbers, no larger than 4 inches high shall be placed on both sides and rear of the vehicle as specified in MIL-STD-642. Instructional markings, such as fuel type and level, shall be marked in letters no larger than one inch high. Safety markings shall be lusterless white, color chip number 37835 and the fire extinguisher pull handle shall be lusterless red, color chip number 31136 as specified in FED-STD-595 (see 4.8.2).

3.6.1 Name, shipping and service data plates. Data plates shall meet the requirements of MIL-P-514, or applicable drawings (see 4.8.2).

3.7 Workmanship. Workmanship shall be of such quality as to assure that the vehicle and its components are free of defects that compromise, limit or reduce capability in the performance of its intended use (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in this specification and additional inspections where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written quality assurance system plan are being properly applied, and (b) Government product inspection to measure the quality of the product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or instances of poor practices which might have an adverse effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been made, or until the conformance of the product to prescribed criteria has been demonstrated.

4.1.4 Qualified products. When a part or component is specified to conform to a specification having a QPL, the contractor shall make available to the Government documentation of the item acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order and quantity.

4.1.5 Quality assurance provisions. In the conduct of inspections, the contractor shall adhere to Quality Assurance Provisions (QAP) and General Quality Assurance Requirements (STA Form 458) as applicable and as required by the documents forming part of this specification.

4.1.6 Certification. Where certification (see 4.8.1) is required to verify material or component conformance to a specification, the contractor shall furnish such certification along with documented test results and performance and analytical data, as applicable.

4.2 Classification of inspection:

a. First article inspection (see 4.3).
   1. First overhauled vehicle inspection (FOVI) (see 4.3.1).
   2. Initial production test (IPT) (see 4.3.2).

b. Quality conformance inspections (QCI) (see 4.4).
c. Control tests (see 4.5).
d. Comparison tests (see 4.6).

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4.3 First article inspection. First article inspection shall be performed on the first overhauled vehicle and on additional vehicle(s), randomly selected from the first month's production, or one from the first 10 produced which shall be designated as initial vehicle. Approval of the first article samples shall not relieve the contractor of the obligation to supply vehicles that are fully representative of those inspected as a first article sample. Any change or deviation of overhauled vehicles from the first article sample shall be subject to the approval of the contracting officer.

4.3.1 FOVI.

4.3.1.1 In-process inspection. The first overhauled vehicle shall be inspected during fabrication to determine conformance of materials and workmanship to specified requirements. These inspections shall preceed application of primer and paint. Processing and welding procedures, the quality system, inspection records, calibration procedures, radiographic procedures and welder certifications shall be reviewed and evaluated.

4.3.1.2 Completed vehicle inspections.

4.3.1.2.1 Contractor inspection. The contractor shall subject the vehicle to a road test and the FOVI specified in table III. The road test shall be performed in accordance with 4.4.4.

4.3.1.2.2 Preliminary inspection. The responsible Government inspection element shall conduct the FOVI specified in table III to the extent that acceptability of the vehicle is verified.

4.3.1.2.3 Provisional inspection. The Government procuring activity shall perform the FOVI specified in table III to the degree required to justify approval of the vehicle.

4.3.1.2.4 Vehicle disposition. After completion of inspection, the first overhauled vehicle shall remain at the manufacturing facility, as a production sample, and be the last vehicle of its type shipped on the contract. The vehicle(s) may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the vehicle(s) during this period in accordance with applicable documents for care and preservation while in storage. All configuration changes taking place after the first overhauled vehicle inspection, shall be made to the first overhauled vehicle(s) so that the vehicle(s) will be representative of the current configuration throughout the life of the contract. No configuration changes may be implemented on subsequent overhauled vehicles after Government conditional acceptance of the first overhauled vehicle (applicable vehicle type) without written authorization from the Government.

4.3.2 IPT.

4.3.2.1 Contractor inspection. The contractor shall examine the IPT vehicle(s) as specified in 4.4.

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TABLE III. Classification of inspections.

Title	Requirements	Inspection	First article		1/ QCI	1/ Control	2/ Comparison
			1/ FOVI	2/ IPT			
Materials and construction	3.2 thru 3.3.3	4.8.1	X	X			X
Defects (see table V)	3/	4.8.2	X	X	X		X
Break-in run	3.3.4	4.8.3	X	X	X		X
Reliability	3.3.5	4.8.4	X	X			X
Maintainability	3.3.6	4.8.5	X	X			X
Durability	3.3.7	4.8.6	X	X			X
Sealer	3.3.8.1	4.8.7		X			X
Bearing seals	3.3.8.2	4.8.7	X	X	X		X
Cab and tailgate seals	3.3.9	4.8.7		X			X
Material handling kit	3.3.10.1	4.8.8	X	X	X		X
Machine gun kit	3.3.10.2	4.8.8		X	X		
Winterization equipment	3.3.10.3	4.8.8		X			X
Winterization kit	3.3.10.4	4.8.8		X			X
Driver's cab cover kit	3.3.10.5	4.8.8		X			X
Personnel heater kit: primary	3.3.10.6	4.8.8		X			X
secondary	3.3.10.7	4.8.8		X			X
Cargo cover kit	3.3.10.8	4.8.8		X			X
Cargo heater kit: primary	3.3.10.9	4.8.8		X			X
secondary	3.3.10.10	4.8.8		X			X
Coolant heater	3.3.10.11	4.8.8		X			X
Air brake kit	3.3.10.12	4.8.8		X			X
Doors, covers and windows	3.3.11	4.8.9	X	X	X	X	X
Engine	3.3.12	4.8.10	X	X	X		X
Engine air induction	3.3.12.1	4.8.11		X	X		X
Exhaust system	3.3.12.2	4.8.12	X	X		X	X
Fuel system	3.3.13	4.8.13	X	X	X	X	X
Fuel system hardware	3.3.13.1	4.8.14		X			X
Fuel shut-off valve	3.3.13.2	4.8.15		X			X
Rack linkage	3.3.13.3	4.8.16	X	X	X		X
Transmission, transfer, differential	3.3.14	4.8.17	X	X	X		
Generating system	3.3.15.1	4.8.18.1	X	X	X		X

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TABLE III. Classification of inspections - Continued.

Title	Requirements	Inspection	First article		1/ QCI	1/ Control	2/ Comparison
			1/ FOVI	2/ IPT			
Lights	3.3.15.2	4.8.18.2	X	X	X		X
Communication circuits	3.3.15.3	4.8.18.3	X	X	X	X	X
Voltage characteristics	3.3.15.4	4.8.18.4	X	X			X
Electromagnetic interference	3.3.15.5	4.8.18.5		X			X
Heaters	3.3.15.6	4.8.18.6		X	X		X
Controls	3.3.16	4.8.19	X	X	X		X
Adjustment mechanisms	3.3.17	4.8.20	X	X	X		X
Stowed material	3.3.18	4.8.21	X	X		X	X
Welding repairs	3.3.19	4.8.22		X	X		X
Blast covers	3.3.20	4.8.23	X	X		X	X
Tailgate	3.3.21	4.8.24	X	X		X	X
Fire extinguisher	3.3.22	4.8.25	X	X		X	X
Environmental conditions	3.4.1	4.8.26.1		X			X
Power package, power train	3.4.2	4.8.26.2		X			X
Engine cooling	3.4.3.1	4.8.26.3		X			X
Transmission lubricant cooling	3.4.3.2	4.8.26.3		X			
Differential oil cooler	3.4.3.3	4.8.26.3		X			
Bilge pumps	3.4.4	4.8.26.4	X	X	X	X	X
Speeds, vehicle control	3.4.5	4.8.26.5	X	X	X		X
Acceleration	3.4.6	4.8.26.6		X	X		X
Stopping	3.4.7	4.8.26.7	X	X	X		X
Turning	3.4.8	4.8.26.8	X	X	X		X
Shallow water fording	3.4.9.1	4.8.26.9	X	X	X		X
Amphibious operation	3.4.9.2	4.8.26.10	X	X	X		X
Trench crossing	3.4.10	4.8.26.11	X	X			X
Vertical obstacles	3.4.11	4.8.26.12	X	X			X
Grade and slope operation	3.4.12	4.8.26.13 4.8.26.14	X	X			X
Grade operation	3.4.12.1	4.8.26.13	X	X			X
Side slope operation	3.4.12.2	4.8.26.14	X	X			X
Suspension	3.4.13	4.8.26.15		X			

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TABLE III. Classification of inspections - Continued.

Title	Requirements	Inspection	First article 1/ FOVI	First article 2/ IPT	1/ QCI	1/ Control	2/ Comparison
Winch system	3.4.14	4.8.26.16	X	X		X	
Winch operation	3.4.14.1	4.8.26.16		X	X		
Winch structural integrity	3.4.14.2	4.8.26.16	X	X		X	
Winch controls	3.4.14.3	4.8.26.16		X	X		
Noise level	3.4.15	4.8.26.17		X			X
Noise hazard sign	3.4.15.1	4.8.26.17		X	X		

1/ At place of overhaul.
2/ At Government proving ground.
3/ Defects requirements: 3.3.4.1, 3.3.9, 3.3.11 thru 3.3.14, 3.3.15.3, 3.3.16, 3.3.17, 3.3.22, 3.4.3.1, 3.4.4, 3.4.13, 3.5, 3.6, 3.6.1, and 3.7.

4.3.2.2 Government inspection. The Government will perform the IPT specified in table III. At least 2 selected vehicles inspected in accordance with 4.5 shall be subjected to a 2000 mile RAM-D test in accordance with table IV at a site(s) selected by the Government. The contractor shall expeditiously furnish repair parts, as required, to adequately support the tests. Quantity of test vehicles and configurations to be tested shall be delineated in the overhaul contract (see 6.2). Unless otherwise specified by the Government procuring activity, the contractor shall refurbish the vehicles after test completion. See 4.8.4 through 4.8.6 for RAM-D requirements.

TABLE IV. 2000 mile overhauled vehicle test with rated payload.

Test course	Mileage and speeds
Hard-surfaced or gravel roads 1/	200 miles at varying speeds up to maximum
Gravel and dirt roads	500 miles at varying speeds up to maximum
Level cross-country	650 miles at varying speeds up to maximum
Hilly cross-country 1/	650 miles at varying speeds up to maximum
Amphibious operation	1 hour minimum at varying speeds up to maximum, type III only

1/ Includes 10 miles with towed load.

4.3.3 Deficiencies. Unresolved failures or deficiencies during, or as a result of, first article inspection shall be cause for rejection of the vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found

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during, or as a result of the IPT shall be evidence that all vehicles already accepted prior to completion of the IPT are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all vehicles shall be corrected by the contractor.

4.4 QCI. A QCI shall be performed in the following sequence and utilizing an approved Final Inspection Record (FIR).

a. Examination for the defects specified in table III and classified in table V.
b. A road test of 8 miles in accordance with 4.4.4.
c. QCI specified in table III.

4.4.1 One hundred percent. Each vehicle shall be subjected to the QCI specified in 4.4.

4.4.2 Acceptable quality level. When authorized by the Government contracting authority, 100% QCI (see 4.4.1) shall be waived. Thereafter, samples shall be selected in accordance with 4.4.2.1.2 and shall be inspected to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	25/100 Units
Minor	100/100 Units

TABLE V. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 25 defects/100 units	
101	Engine; malfunction, leaks, improper mounting (see 3.3.12 and 6.5c).	Visual and functional
102	Transmission; malfunction, leaks, improper mounting (see 3.3.14 and 6.5c).	Visual and functional
103	Transfer gear, differential steer, and final drive; malfunction, leaks (see 3.3.14 and 6.5c).	Visual and functional
104	Fuel system components; malfunction, leaks (see 3.3.13 and 6.5a). 1/	Visual and functional
105	Cooling system components; malfunction, leaks damage (see 3.4.3.1 and 6.5b).	Visual and functional
106	Instrumentation switches, warning and indicating safety devices; malfunction, damage	Visual and functional
107	Electrical system components and circuits, communication system; malfunction (see 3.3.15.3). 2/	Visual and functional
108	Suspension system components and adjustments; damage, leaks, improper clearance, misalignment (see 3.4.13 and 6.5c).	Visual and functional

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TABLE V. Classification of defects - Continued.

Category	Defect	Method of examination
109	Tracks; damage, pad separation (see 3.4.13).	Visual and functional
110	Doors, hatches, cab and tailgate seals, covers and windows; improper fit, malfunction, damage (see 3.3.9 and 3.3.11).	Visual and functional
111	Vision devices and receptacles; damage, malfunction.	Visual and functional
112	Bilge pumps; malfunction, damage (see 3.4.4).	Functional
113	Covers, grilles, ducts, vents, panels, stowage boxes, seats, fenders, hardware; malfunction, locking devices, damage (see 3.3.11).	Visual and functional
114	Controls; malfunction, clearance improper (see 3.3.16).	Visual and functional
115	Track drive sprockets and drive shafts; damage, clearance improper (see 3.4.13).	Visual and SIE 3/
116	Adjustment mechanisms; malfunction, clearance improper (see 3.3.17).	Visual and functional
117	Hull; structural and welding defects.	Visual and functional
118	Towing pintle, lifting eyes, hooks and brackets; malfunction, defects.	Visual and functional
119	Seats, crash pads, and straps; damaged, seat control malfunction.	Visual and functional
120	Fire extinguisher system control; seals missing, components damage (see 3.3.22).	Visual
121	Wiring, tubing or hoses; defective, damage (see 3.3.15.3).	Visual
122	Faulty workmanship affecting performance (see 3.7).	Visual
Minor	AQL 100 defects/100 units	
201	Engine; improper component assembly (see 3.3.12).	Visual and functional
202	Transmission; improper component assembly or installation (see 3.3.14).	Visual and functional
203	Transfer gear, differential steer and final drive units; improper component assembly or installation (see 3.3.14).	Visual and functional
204	Fuel system components; improper line clearance, assembly or installation (see 3.3.13). 1/	Visual and functional
205	Cooling system components; improper assembly or installation (see 3.4.3.1).	Visual and functional
206	Instrumentation, switches, warning, indicating and safety devices; improper assembly or installation.	Visual and functional

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TABLE V. Classification of defects - Continued.

Category	Defect	Method of examination
207	Electrical system components and circuits, communication system; improper assembly or installation (see 3.3.15.3). 2/	Visual and functional
208	Suspension, system components and adjustments; improper installation or assembly, bolt or nut torque (see 3.4.13).	Visual and functional SIE 3/
209	Tracks; tension, installation or assembly improper, including bolt torque, damage (see 3.4.13).	Visual and SIE 3/
210	Doors, hatches, cab and tailgate seals, covers and windows; improper assembly or installation (see 3.3.9 and 3.3.11).	Visual and functional
211	Vision devices and receptacles; improper assembly, installation or clearance.	Visual and functional
212	Bilge pumps; improper assembly, installation (see 3.4.4).	Functional
213	Covers, grilles, ducts, vents, panels, stowage boxes, seats, fenders, hardware; improper fits, adjustment, installation or defective welds, seals, adhesion, hardware or sheet metal (see 3.3.11).	Visual and functional
214	Controls; improper assembly, or installation (see 3.3.16).	Visual and SIE 3/
215	Track drive sprockets and drive shafts; improper assembly, installation or bolt torque (see 3.4.13).	Visual and SIE 3/
216	Hull; improper assembly or installation.	Visual and functional
217	Adjustment mechanisms; improper assembly, or installation (see 3.3.17).	Visual and functional
218	Towing pintle, lifting eyes, hooks and brackets; improper fit, installation, assembly or adhesion, welding defects, damaged.	Visual and functional
219	Seats, crash pads, and straps; data card incomplete, improper assembly or installation.	Visual
220	Fire extinguisher system control; data card incomplete, improper assembly or installation, coding or protection (see 3.3.22).	Visual
221	Wiring, tubing or hoses; defective, missing, improper installation, coding or protection (see 3.3.15.3).	Visual
222	Paint; application or color improper (see 3.5).	Visual
223	Tube fittings; defective, missing, improperly installed (see 3.3.13 and 3.4.3.1).	Visual
224	Decals, marking data and instruction plates; missing, improper location or size, defective (see 3.6 and 3.6.1).	Visual

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TABLE V. Classification of defects - Continued.

Category	Defect	Method of examination
225	Lubrication and lubricants; improper application, types and levels (see 3.3.4.1).	Visual
226	Protective anti-skid coatings; application, materials and coverage improper.	Visual
227	Record forms, and publications; missing, improperly processed or secured.	Visual
228	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ Fuel leaks constitute a special defect and will result in total inspection of the lot for this defect.
2/ Filament failures are not to be classified as major or minor defects.
3/ SIE = Standard Inspection Equipment.

4.4.2.1 Sampling.

4.4.2.1.1 Lot formation. An inspection lot shall consist of all the vehicles overhauled in one month submitted at one time for acceptance.

4.4.2.1.2 Sampling for inspection. When sampling is authorized for QCI (see 4.4.2) samples shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall examine at least 20 consecutively overhauled vehicles. The examination shall establish that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQL.

4.4.3 Defects. The following rules shall be applied to the counting of defects (see 6.3).

4.4.3.1 Unclassified defects. All defects that have no effect on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a thirty-day period, will be added to the minor defects classification with no increase in AQLs.

4.4.3.2 Recurring major deficiencies. A major deficiency is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected by the contractor when found.

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4.4.3.3 Recurring minor deficiencies. A minor deficiency is recurring if it occurs more than twice in the same sample, or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies and correction shall be accomplished prior to acceptance.

4.4.3.4 Leakage defects. The following items constitute a part of the classification of defects (see 6.5):

a. Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
b. Any leak that occurs at a static fit, metal to metal or gasket combination shall constitute a major defect.

4.4.4 Road test. A road test for each overhauled vehicle without payload, shall be conducted for a distance of not less than 8 miles by the contractor. The vehicle shall be completely assembled and serviced. Performance of vehicles shall be demonstrated on smooth, relatively level hard-surfaced roads, and shall meet all requirements specified herein. After completion of the 8 mile road test, the vehicles shall be examined for lubrication leakage and other deficiencies.

4.5 Control tests. The Government shall select at random, after the first production month, one vehicle produced per month. These shall be subjected to control tests in accordance with table III and the following road test.

4.5.1 Fifty mile test. Each overhauled vehicle selected for control tests by the Government shall be operated for a distance of not less than 50 miles by the contractor. Vehicles shall be completely assembled and serviced. The control test vehicles will be weighed with basic issue items (BII). Test courses or areas shall be selected by the manufacturer and shall be on smooth, approximately level, hard-surfaced roads. Vehicles with rated payload shall satisfactorily pass the performance requirements specified herein.

4.6 Comparison tests. The Government may randomly select overhauled vehicles at any time during the production contract period and subject these vehicles to the comparison tests specified in table III and in accordance with the 2000 mile overhaul test of table IV. Tests shall be conducted by the Government at a site it shall select. Vehicles selected for comparison tests shall not have been previously tested according to 4.5, but shall have passed inspection according to 4.4.

4.7 Failure. Failure of any overhauled vehicle to pass any of the specified inspections or failure to meet inspection AQL levels, as applicable, shall be cause for the Government to refuse acceptance of the overhauled quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

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4.8 Method of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports and rating data.

4.8.2 Defects. Conformance to 3.3.4.1, 3.3.9, 3.3.11 through 3.3.14, 3.3.15.3, 3.3.16, 3.3.17, 3.3.22, 3.4.3.1, 3.4.4, 3.4.13, 3.5, 3.6, 3.6.1, and 3.7 shall be determined by examination for the defects listed in table V. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Break-in run (track). To determine conformance to 3.3.4 through 3.3.4.3.3, each vehicle shall be lubricated and serviced prior to the break-in run. The run shall be on smooth, level, hard-surfaced roads, and for the distance specified in table I. The vehicle shall be operated in reverse after each division as specified in 3.3.4.3.1. The undulating portion of the test shall be accomplished in both directions at speeds between 5 and 10 mph during division a. and between 11 and 14.5 mph during division b.

4.8.3.1 Simulated break-in run. To determine conformance to 3.3.4 through 3.3.4.3.3, the simulated break-in run (see table II) may be accomplished on the Army skid-type dynamometer, in lieu of the run specified in 4.8.3. Prior to the break-in run each vehicle shall be lubricated and serviced as specified in 3.3.4.1. Records shall be kept of vehicle instrument readings. The final 10 miles of the total break-in run shall be on smooth, level, hard-surfaced roads. During this 10 miles, the vehicle shall be subjected to undulating terrain operation, in accordance with 3.3.4.3, by being operated over the section of road conforming to figure 1. After completion of divisions c. and d. of table II, the vehicle shall be operated in reverse as specified in 3.3.4.3.1. The undulating portion of the test shall be accomplished twice in each direction at speeds between 5 and 10 mph during division c. and between 11 and 14.5 mph during division d. The results of the break-in run on the first twenty vehicles, that meet all requirements, will serve as a standard for the evaluation of the balance of production.

4.8.3.2 Condition after run. After the break-in run, the vehicle shall be inspected to determine conformance to 3.3.4.3.3.

4.8.4 Reliability verification. To determine conformance to 3.3.5, a point estimate value equal to or greater than 675 miles shall be demonstrated during test. The point estimate value shall be computed by dividing the total cumulative test mileage of all the test vehicles by the total cumulative chargeable failures experienced on these vehicles.

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4.8.5 Maintainability verification. To determine conformance to 3.3.6, a MR equal to or less than that specified in 3.3.6, must be demonstrated during test. The MR will be based upon the total cumulative maintenance manhours with the exceptions of human induced errors and driver or crew checks and services.

4.8.6 Durability verification. Conformance to 3.3.7 shall be verified during test at or above a 50 percent confidence level. The durability requirement will be calculated using a binominal approach requiring a minimum test vehicle sample of two vehicles.

4.8.7 Sealing and seals. To determine conformance to 3.3.8 and 3.3.9 prior to, and after, fording and amphibious operations, the vehicle shall be examined for the amount of water in vehicle and final drives, and lubricant leakage.

4.8.8 Special kits. To determine conformance to 3.3.10 through 3.3.10.12, kits, when specified and installed, shall be operated and examined for proper installation, completeness and function. They shall meet the requirements of 3.3.10.1 through 3.3.10.12.

4.8.9 Doors, windows and covers. To determine conformance to 3.3.11, all doors, windows and covers (including those of special kits when installed) shall be operated and examined for functional requirements. They shall be tested to assure resistance to water, dust and snow.

4.8.10 Engine. To determine conformance to 3.3.12, the engine shall be operated throughout all speed ranges, and examined for proper installation and absence of leakage of lubricants, fuel and coolants. Prior to assembly and installation, all coolant, lubricant, and fuel lines shall be examined to insure internal cleanliness.

4.8.11 Engine air induction system. To determine conformance to 3.3.12.1, the engine air induction system shall be examined for proper installation and absence of leaks.

4.8.12 Exhaust system. To determine conformance to 3.3.12.2, the exhaust system shall be examined for proper installation and absence of leaks. Toxic fume levels measured in the driver and crew compartment, shall not exceed specified values. For the test, the vehicle shall be sealed, the engine operated at 900 to 1000 rpm and the personnel heater, if installed, operated at high heat, for a period of not less than 30 minutes.

4.8.13 Fuel system grades and slopes. With the vehicle operating on specified grades and slopes, the vehicle engine and fuel system shall be observed to determine conformance to 3.3.13.

4.8.14 Fuel system hardware. To determine conformance to 3.3.13.1, the fuel tank shall be examined for cleanliness. The fuel tank(s) filled to capacity, and all fuel lines and connections, shall be examined for leakage before and after all road tests.

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4.8.15 Fuel shut-off valve. To determine conformance to 3.3.13.2, with the engine operating at 750 to 900 rpm, the fuel shut-off handle shall be moved to the off position, and the time for the engine to stop shall be noted.

4.8.16 Rack linkage. To determine conformance to 3.3.13.3, the accelerator pedal shall be operated to full throttle, and the rack examined to assure that full open position has been reached when the pedal is in contact with the stop.

4.8.17 Transmission, transfer gearcase and controlled differential. To determine conformance to 3.3.14, transmission and controlled differential shall be operated throughout all gear ranges, and examined for proper installation and absence of leaks.

4.8.18 Electrical systems.

4.8.18.1 Generating system voltage (alternator). To determine conformance to 3.3.15.1, with the alternator operating as specified, a voltage measurement shall be made with a voltmeter after the described load is removed, and immediately after the road test.

4.8.18.2 Lights. To determine conformance to 3.3.15.2, all lights shall be operated during and after the break-in run throughout all stages of operation, and observed for functional requirements.

4.8.18.3 Communication and electrical circuits. To determine conformance to 3.3.15.3, electrical circuits supplying current to vehicle equipment shall be tested at all points to assure continuity throughout the circuits. Previously tested circuits shall be functionally tested after installation.

4.8.18.4 System voltage characteristics. To determine conformance to 3.3.15.4, system voltage characteristics shall be evaluated as specified in MIL-STD-1275.

4.8.18.5 Electromagnetic interference suppression. To determine conformance to 3.3.15.5, a vehicle shall be tested as specified in MIL-STD-462.

4.8.18.6 Heaters. To determine conformance to 3.3.15.6, verify that the personnel and cargo area heaters start and operate satisfactorily when 20 V dc is applied at the appropriate connectors.

4.8.19 Controls. To determine conformance to 3.3.16, all controls shall be operated as specified and observed for functional requirements.

4.8.20 Adjustment mechanisms. To determine conformance to 3.3.17, all adjustment mechanisms shall be observed during operation, examined for proper adjustment, and adjusted if required.

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4.8.21 Stowed material. To determine conformance to 3.3.18, all manufacturer and depot installed BII shall be stowed in their designated positions on the vehicle. All interior and exterior items shall fit in the spaces provided. The material shall be removed after the test. The BII shall be of the latest configuration available to the contractor. The contractor shall be permitted to unpackage and repackage equipment used for test (for domestic shipment only).

4.8.22 Welding repairs. To determine conformance to 3.3.19, all welding repairs and burning operations shall be examined to determine that slag has been removed and approved procedures have been followed.

4.8.23 Folding blast covers (Type III). To determine conformance to 3.3.20, folding blast covers shall be installed and checked for proper fit and function.

4.8.24 Tailgate (Type III). To determine conformance to 3.3.21, the hinged tailgate shall be checked for proper installation, fit and function.

4.8.25 Fire extinguisher. To determine conformance to 3.3.22, the fire extinguisher release mechanism shall be test operated. This test should be accomplished before the charged cylinders are connected (see 6.4).

4.8.26 Performance. Conditions for performance shall be as specified in 3.4.

4.8.26.1 Environmental conditions. To determine conformance to 3.4.1, a vehicle and its equipment shall be stored and operated at specified temperatures and observed for performance requirements.

4.8.26.2 Power package and power train. To determine conformance to 3.4.2, the vehicle shall be operated throughout all gear and speed ranges. Power plant, power train, tracks and suspension system, and controls shall be observed for proper function and leaks.

4.8.26.3 Cooling system. To determine conformance to 3.4.3.1, 3.4.3.2, and 3.4.3.3, the vehicle shall be operated at the 125°F ambient temperature. The engine, transmission and differential temperatures shall be measured and recorded at the points specified. The results shall be evaluated to determine acceptance. The radiator pressure cap shall be pressure tested and shall relieve pressure within specified range.

4.8.26.4 Bilge pumps. To determine conformance to 3.4.4, the vehicle shall have sufficient water in the hull to measure the performance of the bilge pump(s). Rate of water removal, motor voltage and discharge head shall be measured.

4.8.26.5 Speeds and vehicle control. To determine conformance to 3.4.5, the vehicle shall be driven at speeds from 2 to 3 mph for not less than 5 minutes and at specified speeds thereafter, and performance observed.

4.8.26.6 Acceleration. To determine conformance to 3.4.6, the vehicle shall be operated as specified and acceleration measured.

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4.8.26.7 Stopping. To determine conformance to 3.4.7, the vehicle shall be driven at the specified speed. Three stops shall be made and the results shall be averaged to determine acceptance.

4.8.26.8 Turning. To determine conformance to 3.4.8, the vehicle shall be operated as specified, turned to the right and left in full 360° turns and performance observed.

4.8.26.9 Shallow water fording. To determine conformance to 3.4.9.1, the vehicle shall be driven into water of specified depth without special equipment. Water accumulation shall not exceed the specified amount after 15 minutes of test.

4.8.26.10 Amphibious operation (Type III, only). To determine conformance to 3.3.9 and 3.4.9.2, the vehicle shall be loaded with rated payload, operated amphibiously and performance observed. The amount of water in the hull and leakage of lubricants from the bearings shall be measured.

4.8.26.11 Trench crossing. To determine conformance to 3.4.10, the vehicle shall be driven at a speed not to exceed 5 mph, without stalling, over trenches as specified. After the test, the vehicle shall be examined for damage.

4.8.26.12 Vertical obstacles. To determine conformance to 3.4.11, the vehicle shall be driven at a speed not to exceed 5 mph, without stalling, over vertical obstacles as specified. After the test, the vehicle shall be examined for damage.

4.8.26.13 Grade operation. To determine conformance to 3.4.12 and 3.4.12.1, the vehicle shall be driven up a 60% grade, parked with brakes locked and engine stopped for not less than the specified time and observed for conformance to braking and starting requirements. The test shall be repeated with the vehicle headed downgrade. The test shall be performed with rated load or with a simulated load of equal weight placed in a position that will not restrict the flow of engine air intake or exhaust.

4.8.26.14 Side slope operation. To determine conformance to 3.4.12 and 3.4.12.2, the vehicle shall be driven on 30% side slopes as specified and observed for conformance to braking and starting requirements.

4.8.26.15 Suspension. To determine conformance to 3.4.13, the vehicle, operating at a speed of 18 to 22 mph, shall cross specified obstacles without bottoming out.

4.8.26.16 Winch operation. To determine conformance to 3.4.14 through 3.4.14.3, the winch shall be operated as specified and performance observed.

4.8.26.17 Noise level. To determine conformance to 3.4.15, the vehicle noise level measured at the location indicated shall not be more than the specified dB, when measured with a sound level meter which meets the requirements of MIL-STD-1474. Driver and passenger compartments shall be examined for signs indicating noise hazard, as specified in 3.4.15.1.

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5. PACKAGING

5.1 Vehicle processing. Unless otherwise specified (see 6.2), vehicles shall be processed for storage and shipment as follows:

a. Types I and II vehicles shall be processed in accordance with MIL-C-62015.
b. Type III vehicles shall be processed in accordance with MIL-C-62035.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for use by the US military services as follows:

a. Types I and II: cargo and ammunition support carriers.
b. Type III: guided missile equipment carriers.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and other identifying data (see 1.2).
c. If first article samples are not required (see 3.1).
d. Drawing title, number and date (see 3.3).
e. If simulated break-in run is authorized (see 3.3.4).
f. If overhaul is required for the following installed kits (see 3.3.10):

(1) Materials handling (see 3.3.10.1).
(2) Machine gun (see 3.3.10.2).
(3) Winterization equipment (see 3.3.10.3).
(4) Winterization (see 3.3.10.4).
(5) Driver's cab cover (see 3.3.10.5).
(6) Personnel heater (primary) (see 3.3.10.6).
(7) Personnel heater (secondary) (see 3.3.10.7).
(8) Insulated cargo cover (see 3.3.10.8).
(9) Cargo heater (primary) (see 3.3.10.9).
(10) Cargo heater (secondary) (see 3.3.10.10).
(11) Engine stand-by coolant heater (3.3.10.11).
(12) Air brake (see 3.3.10.12).

g. If welding repairs are authorized, and extent allowed (see 3.3.19).
h. If responsibility for inspection shall be other than as specified (see 4.1).
i. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
j. If first article inspection is not required (see 4.3).
k. Quantity of test vehicles and configurations to be tested (see 4.3.2.2).
l. If vehicle processing shall be other than as specified (see 5.1).

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6.3 Definitions of recurring major and minor defects. The Government inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but, also, for any other departures from good workmanship. The Government inspector will assure that all deficiencies encountered during the inspection are enumerated on the deficiency sheet for the vehicle. The defects noted on the deficiency sheet shall contain sufficient description to enable the Government inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

6.4 Safety precautions. Caution should be exercised in handling carbon dioxide fire-extinguisher cylinders. They should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during reinstallation to avoid tripping the fire extinguisher control system since physical injury is highly probable (see 4.8.25).

6.5 Classification of leakage defects. The following definitions shall be used with the classification of leakage defects:

a. Weep: Any evidence of fluid beyond a seal or splitline.
b. Seep: Any evidence of fluid beyond a seal or splitline that does not result in formation of a droplet.
c. Leak: Any evidence of fluid beyond a seal or splitline that results in the formation of a droplet.
d. Drip: Any evidence of fluid beyond a seal or splitline where a droplet forms and falls.

6.6 Deficiency sheet. The Government inspector shall verify that a thorough inspection of each vehicle is performed by the contractor for the listed characteristics, and for any departures from good workmanship. The Government inspector shall assure that all defects encountered during inspection are enumerated on a deficiency sheet for the vehicle. Defects noted on a Deficiency Sheet shall contain sufficient description to enable the Government inspector and the contractor's representative to classify the defects in accordance with the classification of defects (see table V) and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring defects.

6.7 Electromagnetic interference suppression. The contracting officer should submit a copy of the reports required (see 3.3.15.5) to Commander, US Army Electronics Research and Development Command, ATTN: DELET-R-S, Fort Monmouth, NJ 07703-5302 for approval.

6.8 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.9 Mission failure scoring criteria. M113A2 Family Vehicles Failure Definition and Scoring Criteria may be obtained from the US Army Tank-Automotive Command, Warren, MI 48397-5000, ATTN: AMSTA-QRA.

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6.10 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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